

SEONGLAE CHO

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Senior AI Engineer at Holistic AI and mechanistic-interpretability researcher. First-author ICML 2026 main publication; agent runtime safety through steering, probe-as-tool, and runtime agent monitoring, taken from research to production.

SELECTED PUBLICATIONS

- Cho, S., Wu, Z., & Koshiyama, A. (2025). [CorrSteer](#): Generation-Time LLM Steering via Correlated Sparse Autoencoder Features. **ICML 2026 main**. +27.2% HarmBench (108 samples), +3.3% MMLU. [demo](#)
- Cho, S., Wu, Z., & Koshiyama, A. (2026). [Control Reinforcement Learning](#): Interpretable Token-Level Steering of LLMs via Sparse Autoencoder Features. ICLR 2026 Workshop on Principled Design for Trustworthy AI.
- Cho, S. et al. (2026). PaaT (Probe as a Tool): self-regulating agents via a callable activation probe. ICML 2026 FAGen Workshop. Probe AUROC 0.93–1.00 across 11 risk heads. [OpenReview](#)
- Cho, S. et al. (2026). AgentRoom: Concurrent Multi-Agent Coding in a CRDT-Backed Shared Workspace. ICML 2026 FAGen Workshop. Task-abandonment reduction holds across 3 models and 4 tasks under a judge-independent classifier. [OpenReview](#)
- Cho, S., Wu, Z. et al. (2026). Automata from Agent Traces: Failure and Next-Step Prediction. ICML 2026 AIWILD Workshop. 0.94 AUROC failure detection. [OpenReview](#)

PUBLICATIONS

- Cho, S., Wu, Z. et al. (2026). [The Confidence Manifold](#): Geometric Structure of Correctness Representations in Language Models. ICLR 2026 Workshop on Agentic AI in the Wild. Internal probes far outperform P(True) and semantic entropy on adversarial misconceptions across 11 models. [OpenReview](#)
- Cho, S., Wu, Z. et al. (2026). Are Single-Token Sparse Autoencoder Features Irreplaceable? The Impact of Training Methodology. EMNLP 2026 under review.
- Cho, S. et al. (2025). [FaithfulSAE](#): Towards Capturing Faithful Features with Sparse Autoencoders without External Datasets Dependency. ACL 2025 SRW. Outperformed external-data SAEs in 5/7 models.
- Wu, Z., Cho, S. et al. (2025). AgentGraph: Trace-to-Graph Platform for Interactive Analysis and Robustness Testing in Agentic AI Systems. AAAI 2026 Demo.
- Wu, Z., Cho, S.* et al. (2025). [LibVulnWatch](#): A Deep Assessment Agent System and Leaderboard for Uncovering Hidden Vulnerabilities in Open-Source AI Libraries. ICML 2025 TAIG Workshop. [demo](#)
- Cho, S. et al. (2024). [RTSUM](#): Relation Triple-based Interpretable Summarization with Multi-level Saliency Visualization. NAACL 2024 System Demonstrations.

EXPERIENCE

HOLISTIC AI, London, UK · *Senior AI Engineer* May 2025 – Present

- Built a trace-based AI agent evaluation framework with structured traces, quality criteria, and graph-based monitoring; a 300MB toxic-chat classifier beat gpt-oss-safeguard-20B on F1 (0.836 vs 0.799).
- Designed a trace-to-FSM reconstruction method reaching 0.904 runtime-monitoring F1 at 68% lower compute.
- Integrated ReBAC access control into the AI Governance platform with database-level policy enforcement.
- Promoted to Senior within one year of joining.

YONSEI UNIVERSITY, Data & Language Intelligence Lab, Seoul, South Korea · *Research Intern, Prof. Dongha Lee* Mar 2023 – Jan 2024

- Interpretable-summarization research on LLMs; first-authored RTSUM (NAACL 2024 Demo).

KAKAO MOBILITY, Seoul, South Korea · *Software Engineer, Digital Twin Team* Dec 2021 – Sep 2022

- Built Kubernetes infrastructure with Grafana alerting and isolated namespaces.
- Led a 3-engineer team delivering a 3D point-cloud processing service handling 10M+ point cloud frames in the autonomous-driving pipeline.
- Ported a C++ 3D-projection algorithm to Rust with Node.js N-API bindings for cross-platform deployment.

STRYX, Seoul, South Korea · *Software Engineer, Geospatial AI Startup* Nov 2019 – Dec 2021

- Built LiDAR point-cloud data pipelines: Potree conversion for web visualization, point-cloud interpolation, and PCD/LAS processing.
- Cut build time 70% by consolidating multiple repositories into a monorepo.
- Reduced the Docker image from 2GB to 180MB via multi-stage CI builds.

EDUCATION

University College London · *MSc in AI for Sustainable Development, Distinction* Sep 2024 – Sep 2025

- Dissertation: Representation-Level Steering Improves Task Performance and Safety in LLMs. Supervisor: Prof. Philip Treleaven. Mentor: Zekun Wu.

Yonsei University · *BE in Computer Science* Mar 2017 – Aug 2024

AWARDS

- HERMES, 1st Place (£3,000), Holistic AI Hackathon** (Nov 2024) · cut biased, stereotypical text generation from 90% to 70% with an SAE-derived steering vector.
- MBTIGPT, 1st Place (₩3,000,000), Yonsei GenAI Competition** (2023) · AI service with 1,000+ users, RAG over chat history with Redis and Faiss.

PROJECTS

- OptimismBench** · multilingual benchmark for directional bias in LLM probability judgment across 6 languages and 16+ models. Feb 2026 – Present
EMNLP 2026 under review.
- Stochastic Grokking** · grokking hinges on a few attention heads binding to variable slots; transplanting these heads rescues 6/6 failed seeds. NeurIPS 2026 under review. Jan 2026 – Present
- RouterRL** · trained only MoE router parameters with dense layer-wise probe rewards, which learn the routing policy where sparse binary RL does not. Feb 2026 – Mar 2026

- **Vect** · local-first multi-agent coding IDE orchestrating multiple coding-CLI agents in a shared workspace.
 - **LLaMa2GPTQ** · local LLM with 4-bit GPTQ quantization (~75% memory) and RAG vector search.
 - **ReSRer** · retrieval-augmented generation pipeline indexing 21M documents into a Milvus vector database.
 - **Texonom** · Zettelkasten knowledge system; ANN retrieval over 30,000 pages embedded into Postgres pgvector.
- Mar 2026 – Present

Jun 2023 – Jul 2023

Sep 2023 – Jan 2024

Nov 2021 – Jun 2023

SKILLS

- **AI:** Python, PyTorch, Hugging Face, vLLM, ONNX, DDP/FSDP, PydanticAI, RAG, LangGraph, Agent SDK, Langfuse, LangSmith.
- **Service:** TypeScript, Node.js, Rust, React, Next.js, PostgreSQL, Redis, Faiss, Milvus, Vite.
- **Infra:** Kubernetes, Linux, Git, CI/CD, Docker, Apache Spark, Ansible, ETL, AWS, GCP, Hadoop, Terraform, SST.
- **Languages:** Python, TypeScript, Rust, C++.

SERVICE

- Reviewer, ICML 2026 (Silver Reviewer Award) · Reviewer, ACL Rolling Review (March 2026).
- Mandatory military service completed as an Industrial Technical Personnel.